

Readme.md

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Overview

This directory contains necessary scripts and data to reproduce the results and figures presented in “*Robust Regression for Spherical Data in an Extrinsic Semiparametric Framework*”. Since the simulation studies are computationally intensive, we provide the simulated data and estimation results in the corresponding .RData files, which will be used to produce relevant figures.

Files in the Directory

- **data**: This folder contains **gemas_merge.csv** for real data application and .RData files that store simulated data and estimation results.
- **plot.R**: this script produces the graphical illustrations of the asymptotic relative efficiency and standardized gross error sensitivity in Section 5. Graphical illustrations of the contours of the distributions on the sphere and shape of link function are also included.
- **ESLFns.R**: this script provides estimation methods for all regression models considered in the numerical studies, including the extrinsic single index model, Fréchet single-index model with extrinsic distance and single-index quantile regression.
- **ESIM_contaminate.R**: this script reproduces simulations for Section 6.1. The relevant simulated data and estimation results are saved in **simu_contaminate_XXX.RData**, which can be used to generate the corresponding figures.
- **ESIM_shape.R**: this script reproduces simulations and relevant figures for Section 6.2. The relevant simulated data and estimation results are saved in **simu_shape.RData**.
- **ESIM_power.R**: this script reproduces simulations for Section 6.3.
- **GEMAS_sand.R**: this script reproduces the data application to **gemas_merge.csv** in Section 6.4, with the estimation results saved in **GEMAS_sand.RData**.
- **ESIM_cov.R** and **heavy_tailed.R**: these scripts reproduce additional numerical results in the supplementary material, with the estimation results saved in **simu_cov_XXX.RData** and **simu_heavy.RData**, respectively.

Key Functions

We provide a brief overview of the main functions in **ESLFns.R** that implement the proposed methodology. A detailed introduction to their implementations can be found in Section S3 of the supplementary material.

- **ls_esl**: This function performs the estimation for the extrinsic single-index model with the least squares.
- **esl_est**: This function performs the estimation for the extrinsic single-index model with the exponential squared loss. It involves iteratively reweighted least squares, and includes estimation of the tuning parameter λ . The function **cost_lambda_median** is used to estimate λ during this iterative process.
- **siqr_est**: This function performs the estimation for the single-index quantile regression with a quantile level of 0.5. It adapts the estimation approach proposed by Wu et al. (2010) and extends to multivariate response.

- **fsim_est**: This function performs the estimation for the single-index Fréchet regression proposed by Bhattacharjee and Müller (2023). For a detailed explanation of the methodology, please refer to Section S3.2 of the supplementary material. The code is adapted from <https://github.com/functionaldata/tFrechet>.
- **simProject**: This function generate random samples following the scaled von Mises Fisher distribution. See Scealy and Wood (2019) for details.

Required R packages:

- Directional (6.9)
- KernSmooth (2.23-24)
- quantreg (5.99)
- frechet (0.3.0)
- MASS (7.3-60.2)

Reference

- Bhattacharjee, S. and Müller, H.-G. (2023). Single-index Fréchet regression. *The Annals of Statistics* **51**.(4), pp. 1770-1798.
- Lin, L., St. Thomas, B., Zhu, H., and Dunson, D. B. (2017). Extrinsic local regression on manifold-valued data. *Journal of the American Statistical Association* **112**.(519), pp. 1261-1273.
- Scealy, J. L. and Wood, A. T. A. (2019). Scaled von Mises-Fisher distributions and regression models for paleomagnetic directional data. *Journal of the American Statistical Association* **114**.(528), pp. 1547-1560.
- Wu, T. Z., Yu, K., and Yu, Y.(2010). Single-index quantile regression. *Journal of Multivariate Analysis* **101**.(7), pp. 1607-1621.